04_js_en

Laboratory 04

Task 1

Node.js - Padel Console Tracker (REPL/CLI).
Create a simple console application to track padel statistics, launched by `node app.js`.

STEP BY STEP:

1. Declare 3 player objects with the following data:

```
{ name: "Marcin", elo: 1567, matches: 124, wins: 47 }
{ name: "Jan", elo: 1423, matches: 89, wins: 34 }
{ name: "Anna", elo: 1689, matches: 156, wins: 102 }
```

2. Write 5 clean functions:
   - calculateWinrate(player) → win rate
   - sortByElo(players) → sort by ELO
   - getTopPlayer(players) → top player
   - addMatch(player, won) → add match
   - displayStats(player) → format statistics
3. Main loop from process.argv :

```
node app.js add-match Marcin true
node app.js top
node app.js list
```

4. Rating: ELO 1000–2500, wins ≤ matches
5. Coloured logs with \x1b[32m

PRZYKŁADOWY EFEKT KOŃCOWY:

```
$ node app.js list
🥇 Marcin (ELO: 1567) | 124 mecze | 62% winrate √
🥈 Anna  (ELO: 1689) | 156 mecze | 65% winrate √
🥉 Jan   (ELO: 1423) | 89 mecze  | 38% winrate
```

Task 2

Browser-based - Interactive Padel Calculator
Create a browser-based calculator for padel match scores (3 sets of 3 games).

STEP BY STEP:

1. Retrieve DOM elements using `querySelector`
2. 4 event handler functions:
   - addPoint(player) → points (11, 15, 21 system)
   - newSet() → new set (0:0)
   - newMatch() → new match
   - updateDisplay() → update UI
3. Event listeners:

```
addPointBtn1.addEventListener('click', () => addPoint('player1'))
document.addEventListener('keydown', handleKeyboard)
```

4. Padel rules: sets to 2 wins, points with a 2-point margin
5. Validation + UI states

Sample HTML template:

```
<!DOCTYPE html>
<html>
<head>
  <title>Padel Score Keeper</title>
  <style>body{font-family:system-ui;padding:2rem}</style>
</head>
<body>
```

```html
  <h1>Padel Score Keeper</h1>
  <div class="scoreboard">
    <div id="player1">Marcin: <span id="player1-score">0</span></div>
    <div id="player2">Opponent: <span id="player2-score">0</span></div>
    <div>Sets: <span id="sets">0:0</span></div>
  </div>
  <button id="add-p1">+1 Marcin</button>
  <button id="add-p2">+1 Opponent</button>
  <button id="new-set">New Set</button>
  <button id="new-match">New Match</button>
</body>
</html>
```

EXAMPLE OF THE FINAL RESULT:

```
Marcin: 8    Opponent: 6    Sets: 1:0
[+1 Marcin] [+1 Opponent] [New Set] [New Match]
```

Keyboard: A/D or ←/→

Solutions

Task 1: app.js (Node.js)

```javascript
const players = [
  { name: "Marcin", elo: 1567, matches: 124, wins: 47 },
  { name: "Jan", elo: 1423, matches: 89, wins: 34 },
  { name: "Anna", elo: 1689, matches: 156, wins: 102 }
];

const calculateWinrate = (player) => Math.round((player.wins / player.matches) * 100);
const sortByElo = (players) => [ ... players].sort((a, b) => b.elo - a.elo);
const getTopPlayer = (players) => sortByElo(players)[0];

```

```
const displayStats = (player) => {
  const winrate = calculateWinrate(player);
  console.log(`🥇 ${player.name.padEnd(8)} (ELO: ${player.elo}) | ${player.matches} mecze | ${winrate}% winrate`);
};

if (process.argv[2] === 'list') {
  sortByElo(players).forEach(displayStats);
} else if (process.argv[2] === 'top') {
  displayStats(getTopPlayer(players));
}
```

Task 2: script.js (Browser)

```
let score1 = 0, score2 = 0, sets1 = 0, sets2 = 0;

const updateDisplay = () => {
  document.getElementById('player1-score').textContent = score1;
  document.getElementById('player2-score').textContent = score2;
  document.getElementById('sets').textContent = `${sets1}:${sets2}`;
};

document.getElementById('add-p1').addEventListener('click', () => {
  score1++;
  if (score1 >= 11 && score1 - score2 >= 2) {
    sets1++; score1 = 0; score2 = 0;
  }
  updateDisplay();
});

document.getElementById('new-match').addEventListener('click', () => {
  score1 = score2 = sets1 = sets2 = 0;
  updateDisplay();
});
```